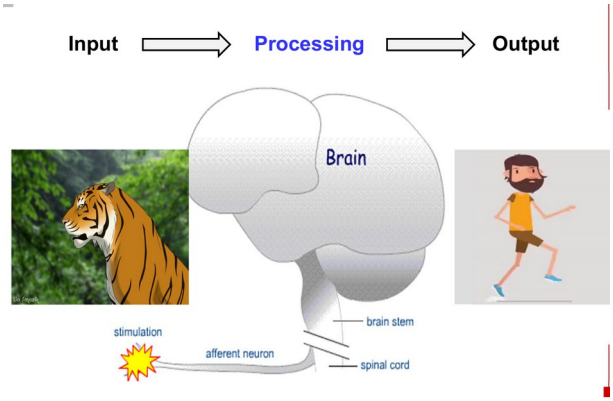
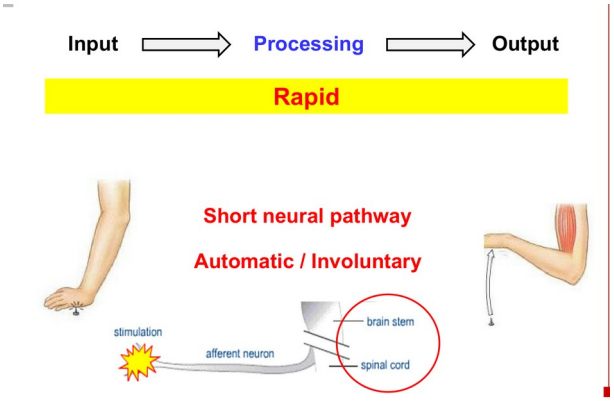
01 Reflexes & the reflex arc

Rapid responses • Defined circuits • Predictable effects

Reflex = a rapid, automatic and predictable reaction to a specific stimulus.



Input → processing → output

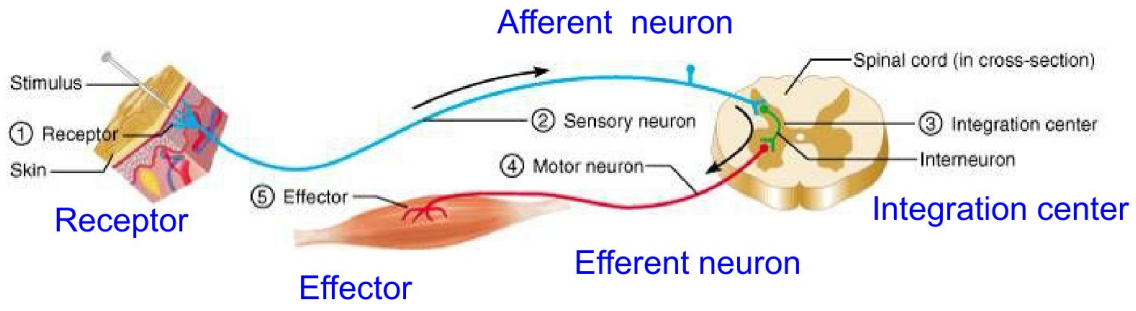


A short neural pathway produces a rapid, involuntary response.

The five components

Receptor → sensory (afferent) neuron → integration centre → motor (efferent) neuron → effector

Component	Role
Receptor	Detects the stimulus.
Sensory neuron	Carries sensory impulses from the receptor toward the CNS.
Integration centre	Processes the input and coordinates the response; usually within the CNS for a typical reflex arc.
Motor neuron	Carries motor commands from the CNS to the effector.
Effector	A muscle or gland produces the response.

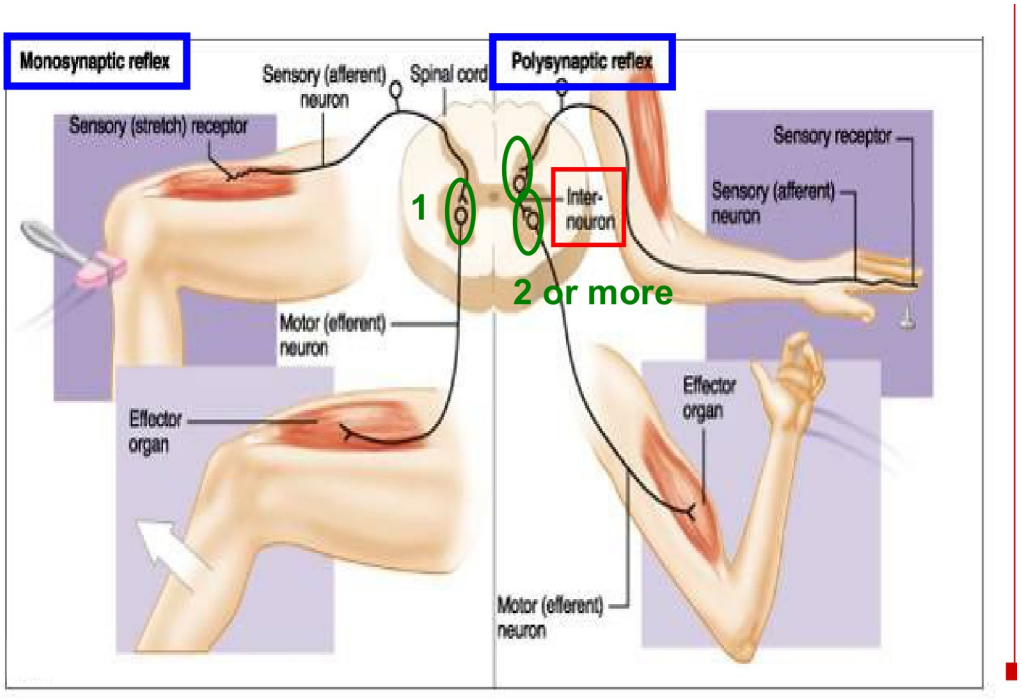


Afferent = toward the integration centre. Efferent = away from it. Short enteric reflexes use a peripheral integration centre.

02 Classifying reflexes

The same reflex can be classified on several independent axes.

Classification	Meaning / example
Innate vs acquired	**Innate:** inborn; withdrawal from pain. **Acquired:** learning-dependent; braking when a red traffic light appears.
Spinal vs cranial	**Spinal:** processing in the spinal cord; withdrawal from pain. **Cranial:** processing in the brain; movement in response to a sudden loud noise.
Somatic vs autonomic	**Somatic:** skeletal muscle effectors. **Autonomic:** smooth muscle, cardiac muscle or gland effectors.
Monosynaptic vs polysynaptic	**Monosynaptic:** one central synapse, directly linking sensory and motor neurons. **Polysynaptic:** two or more central synapses; interneuron(s) intervene.



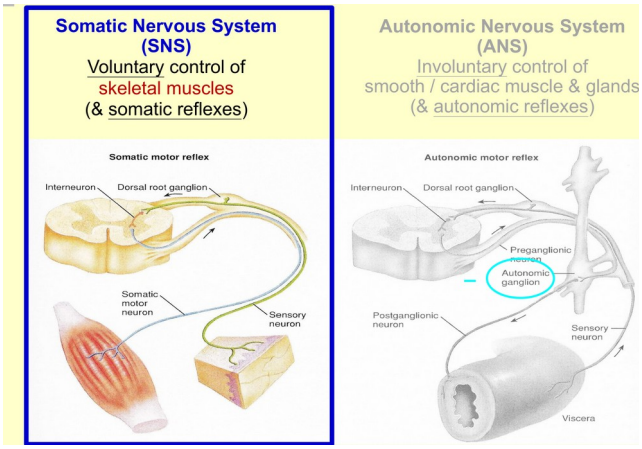
- Count the synapses in the reflex integration pathway, not the number of muscles or the neuromuscular junction.
- A monosynaptic pathway has no intervening interneuron between its sensory neuron and motor neuron.



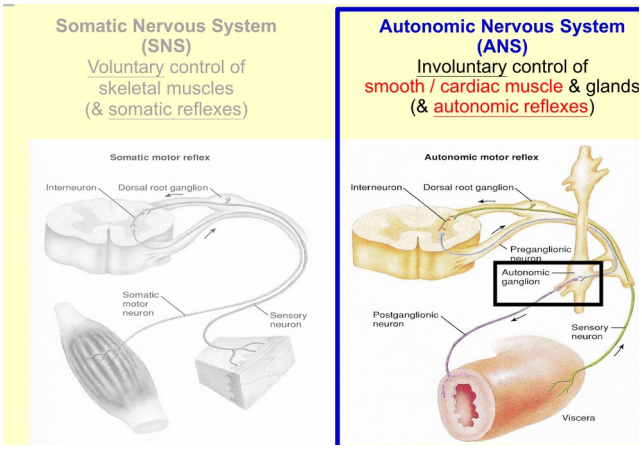
Learned braking and an auditory stimulus are examples used to classify reflexes.

03 Somatic vs autonomic motor output

Feature	Somatic	Autonomic
Effectors	Skeletal muscle	Smooth muscle, cardiac muscle and glands
Control	Voluntary actions and involuntary somatic reflexes	Involuntary visceral control and autonomic reflexes
Efferent pathway	One somatic motor neuron from CNS to skeletal muscle	Preganglionic neuron → autonomic ganglion → postganglionic neuron → effector

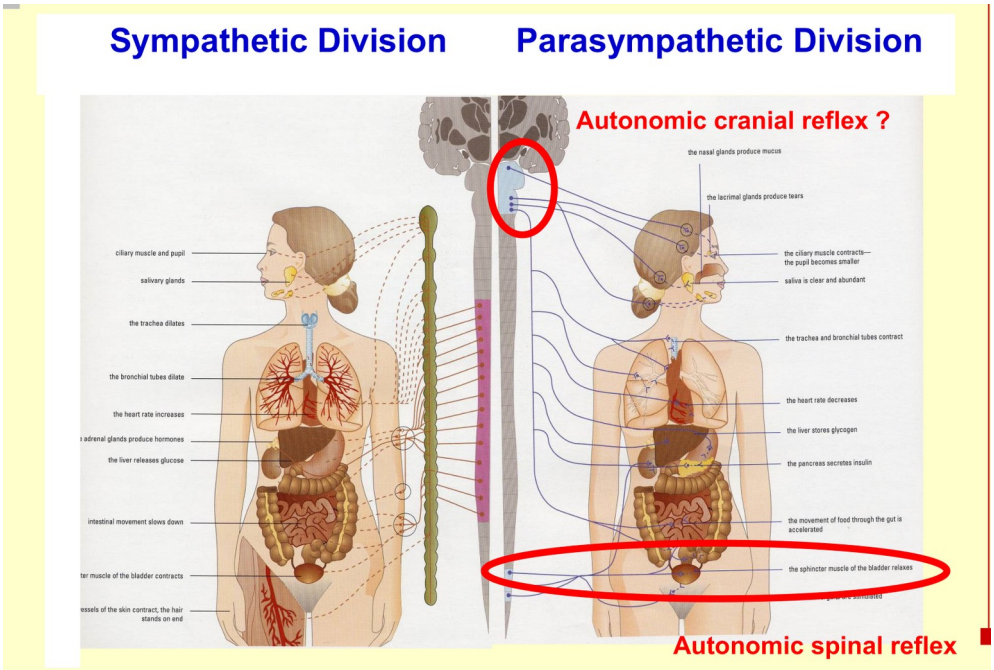


Somatic motor pathway



Autonomic motor pathway

Autonomic organisation

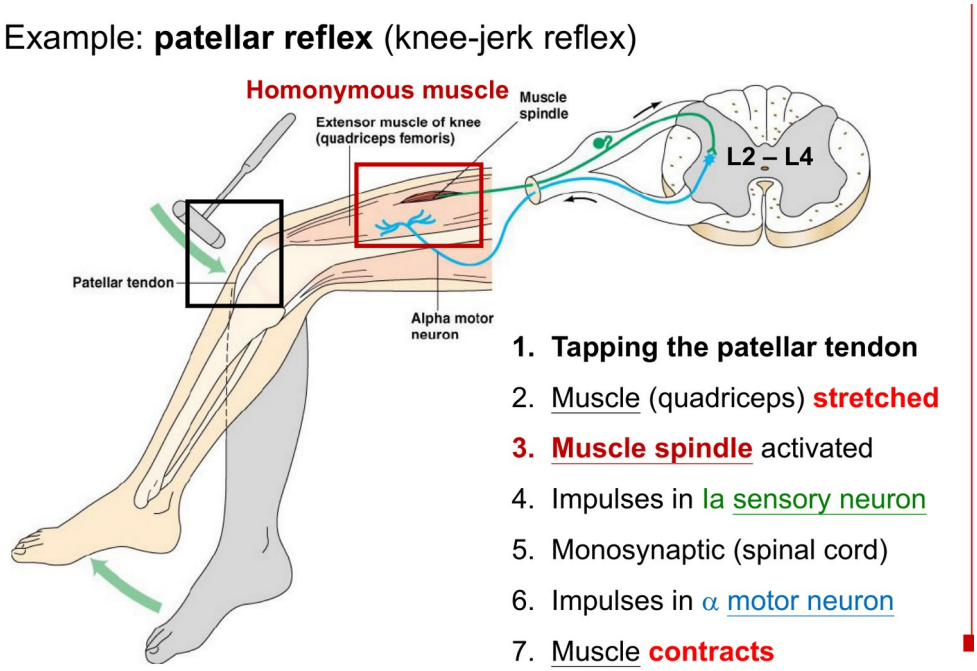


- Parasympathetic output has cranial and sacral components: pupillary light reflex is cranial; defecation and micturition include spinal autonomic circuits.

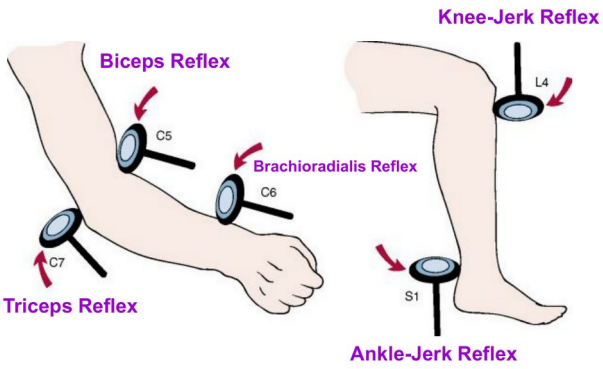
CNS = central nervous system; ANS = autonomic nervous system. Here, “somatic” identifies the skeletal-muscle pathway; it does not mean that every action is voluntary.

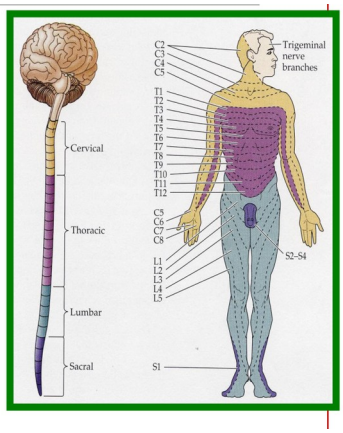
04 Stretch reflex: the knee jerk

Patellar tendon tap → quadriceps stretches → **muscle spindle** activated
→ **Ia sensory firing ↑** → spinal cord (**L2–L4; monosynaptic excitation**)
→ **α-motor neuron firing ↑ → the same quadriceps contracts** → knee extension



- **Homonymous muscle = the same muscle** that contains the stimulated receptor.
- The receptor is the muscle spindle; tapping the tendon mechanically stretches the muscle.





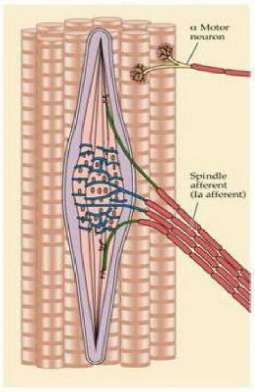
Reflex	Key segment emphasis
Biceps / brachioradialis / triceps	C5 / C6 / C7, respectively
Knee jerk / ankle jerk	L4 emphasis within the L2–L4 knee-jerk circuit / S1 emphasis

Different reflexes use different spinal segments → comparing reflex responses helps localise a lesion or neuropathy.

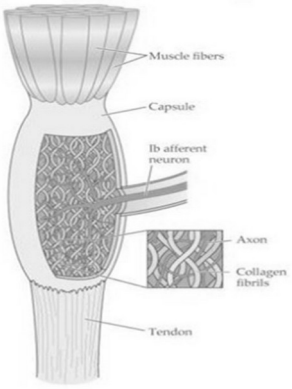
05 Muscle spindle: structure & innervation

Proprioceptors in skeletal muscles:

Muscle spindle
Detects (change in) muscle length
Receptor for **stretch reflex**



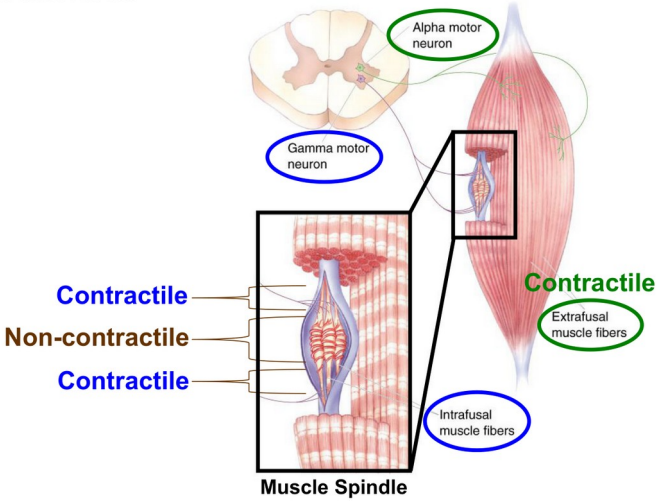
Golgi tendon organ
Detects tension
Receptor for **Golgi tendon reflex**



Structure / neuron	Key role
Muscle spindle	Proprioceptor within skeletal muscle; detects muscle length and changes in length.
Intrafusal muscle fibres	Specialised fibres inside the spindle. Contractile polar ends flank a non-contractile sensory region.
Extrafusal muscle fibres	Ordinary force-producing skeletal muscle fibres outside the spindle.
Ia sensory ending	Reports stretch of the spindle's sensory region to the spinal cord.
α-motor neuron	Innervates extrafusal fibres → produces muscle contraction.
γ-motor neuron	Innervates contractile intrafusal ends → adjusts tension on the spindle's sensory region.

Alpha-Gamma Co-activation

Different Innervations

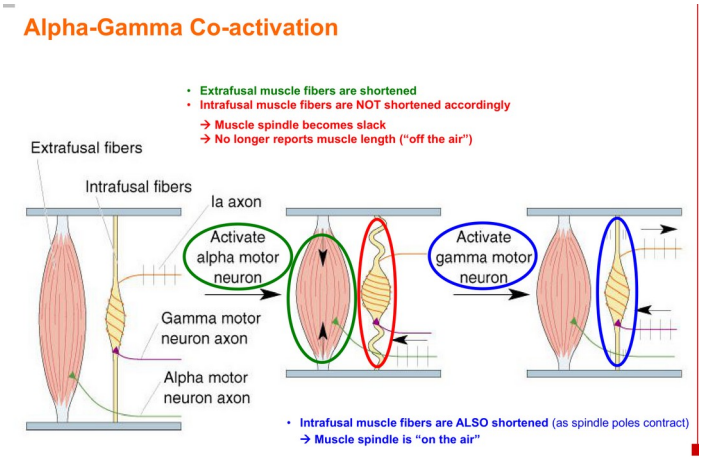


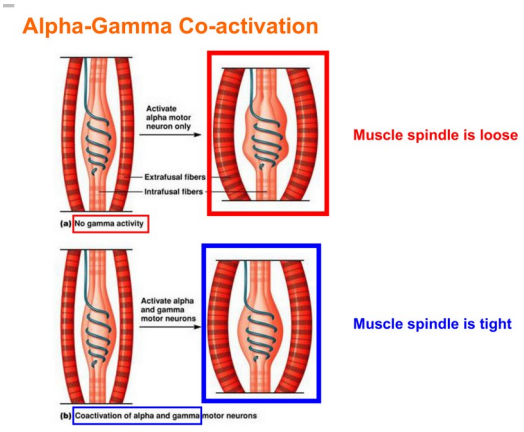
α acts on extrafusal fibres; γ acts on intrafusal fibres. The spindle senses stretch; the Golgi tendon organ senses tension.

06 α–γ coactivation: preserve sensitivity

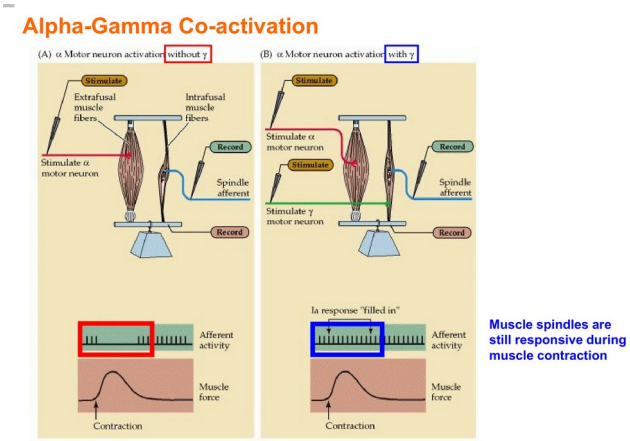
Descending command → **α and γ motor neurons activated together**
α → extrafusal fibres shorten
γ → intrafusal poles contract → central sensory region stays taut
→ **spindle remains responsive during muscle contraction**

α activation alone	α + γ coactivation
Extrafusal fibres shorten.	Extrafusal fibres and intrafusal poles shorten together.
Intrafusal fibres do not shorten accordingly → spindle becomes slack.	Intrafusal polar contraction maintains tension on the sensory region.
Sensory reporting is reduced or lost: spindle is “off the air”.	Length changes can still be detected: spindle remains “on the air”.





Loose vs taut spindle

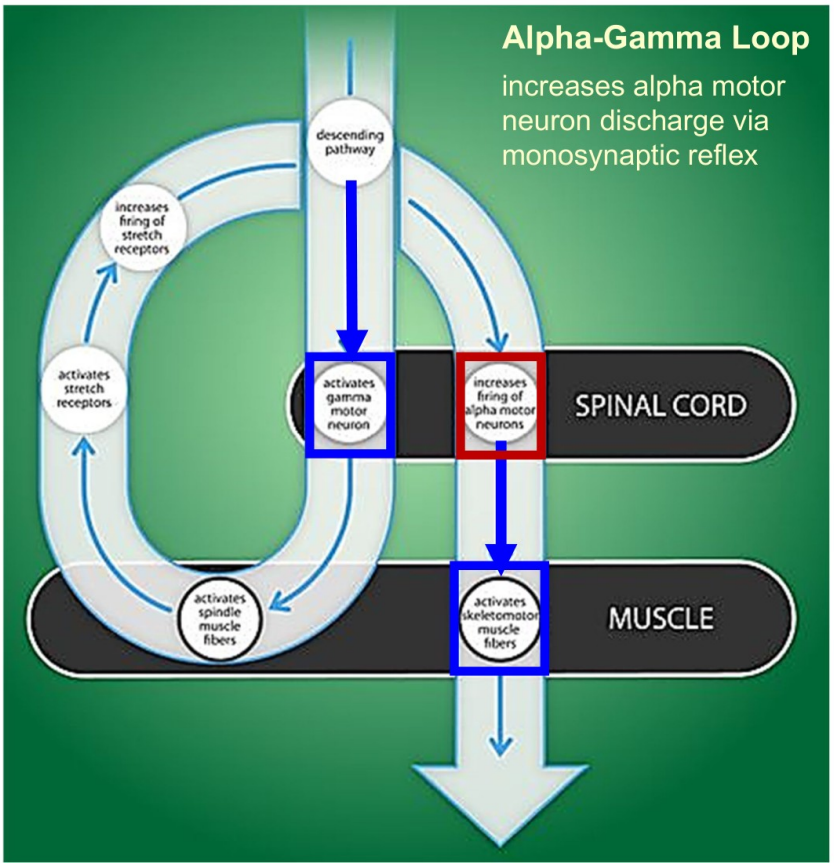


Sensory activity is maintained with coactivation.

- **Purpose: maintain spindle sensitivity while the force-producing muscle shortens.**

07 α–γ loop: generate muscle tone

Descending input → **γ-motor neuron activity ↑**
→ intrafusal polar ends contract → spindle sensory region stretched
→ **Ia afferent firing ↑** → monosynaptic excitation in spinal cord
→ **α-motor neuron firing ↑** → extrafusal contraction → muscle tone



Mechanism	What to remember
α–γ coactivation	α and γ receive descending activation together → spindle sensitivity is maintained during contraction.
α–γ loop	γ activity increases α activity **via the spindle–Ia stretch-reflex pathway** → contributes to muscle tone.

- γ-motor neurons do not directly provide the motor supply to ordinary extrafusal muscle fibres.
- Muscle tone supports posture and readiness for movement.

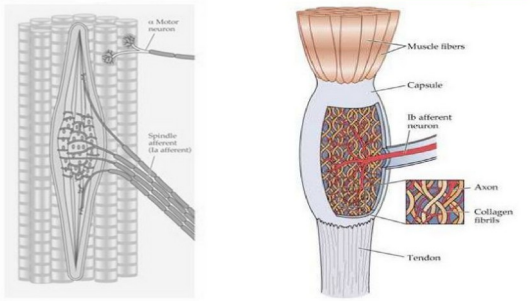
08 Golgi tendon reflex: tension control

Muscle contracts → tension ↑ → **Golgi tendon organ** activated
→ **Ib sensory firing ↑** → spinal interneuron pathway (**polysynaptic**)
→ **α-motor firing to the same muscle ↓ → muscle relaxes**

Proprioceptors in skeletal muscles:

Muscle spindle
Detects (change in) muscle length
Receptor for **stretch reflex**

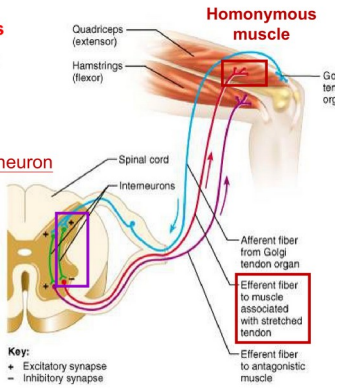
Golgi tendon organ
Detects tension
Receptor for **Golgi tendon reflex**



Tendon organ detects tension.

Example:

1. Muscle (quadriceps) **contracts**
2. **Golgi tendon organ** activated
3. Impulses in Ib sensory neuron
4. Polysynaptic (spinal cord)
5. Reduced impulses in α motor neuron
6. Muscle **relaxes**



Inhibitory control of the homonymous muscle.

Feature	Stretch reflex	Golgi tendon reflex
Stimulus	Muscle length / stretch ↑	Muscle tension ↑
Receptor	Muscle spindle	Golgi tendon organ
Sensory fibre	**Ia**	**Ib**
Main spinal connection	Monosynaptic excitation	Polysynaptic inhibition
Same-muscle response	α firing ↑ → contraction	α firing ↓ → relaxation
Function	Maintains length, posture and tone	Regulates tension; protective role

- The inhibitory interneuron decreases output to the contracting muscle. An excitatory pathway to the antagonist is also shown in the circuit.

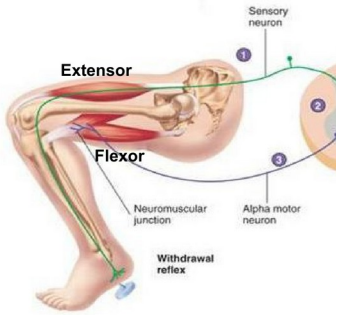
Length → spindle → Ia → contraction.
Tension → tendon organ → Ib → relaxation.

09 Withdrawal & crossed extension

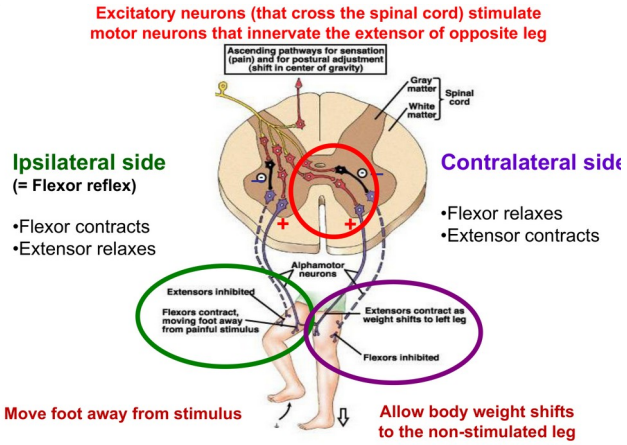
Noxious stimulus (e.g. stepping on a nail) → **pain receptor**
→ pain afferent → spinal cord (**polysynaptic**)
→ ipsilateral flexor α-motor firing ↑ → **flexor contraction → withdrawal**

Example:

1. Foot steps on a nail
2. **Pain receptors** activated
3. Impulses in pain afferent
4. Polysynaptic (spinal cord)
5. Impulses in α motor neuron
6. Flexor muscle **contracts**
7. Withdrawal response



Withdrawal of the stimulated limb



Opposite limb supports the body.

Side	Flexor	Extensor	Outcome
Ipsilateral Stimulated side	Contracts	Relaxes	Foot moves away from the painful stimulus.
Contralateral Opposite side	Relaxes	Contracts	Body weight shifts to the non-stimulated leg.

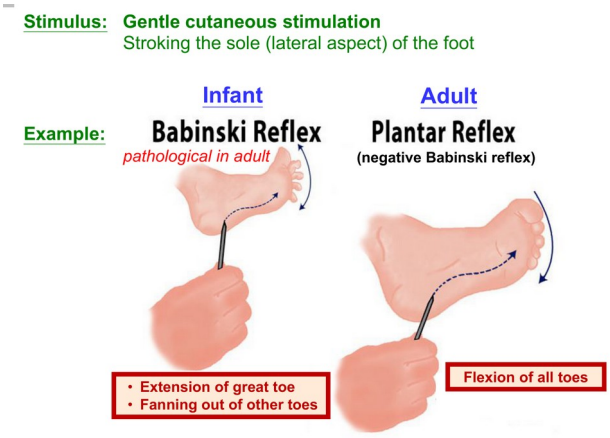
- Ipsilateral reciprocal inhibition relaxes the extensor while the flexor contracts.
- Interneuronal pathways cross the spinal cord → excite motor neurons supplying the opposite extensor and inhibit the opposite flexor.
- Sensory information can also ascend to the brain; spinal processing produces the withdrawal response.

Withdrawal protects the stimulated limb.
Crossed extension maintains balance and posture.

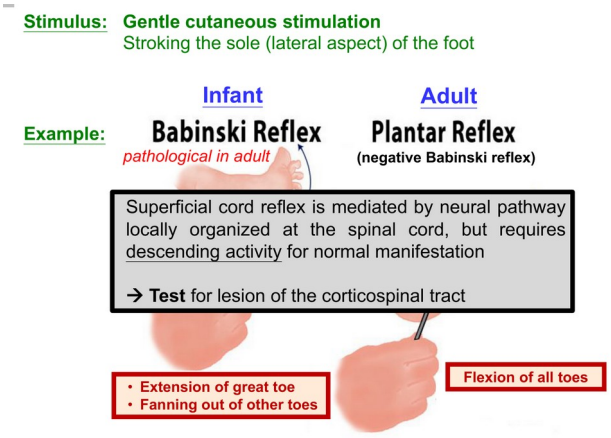
10 Superficial reflex: plantar response

Gentle cutaneous stimulation: **stroke the lateral sole of the foot**
→ sensory input → locally organised spinal circuit → toe response

Pattern	Response	Interpretation
Normal adult plantar response	**Flexion of all toes**	Negative Babinski response.
Positive Babinski response	**Great toe extends; other toes fan out**	Can occur normally in infants; **pathological in adults**.



Compare the adult and infant patterns.



Normal expression requires descending control.

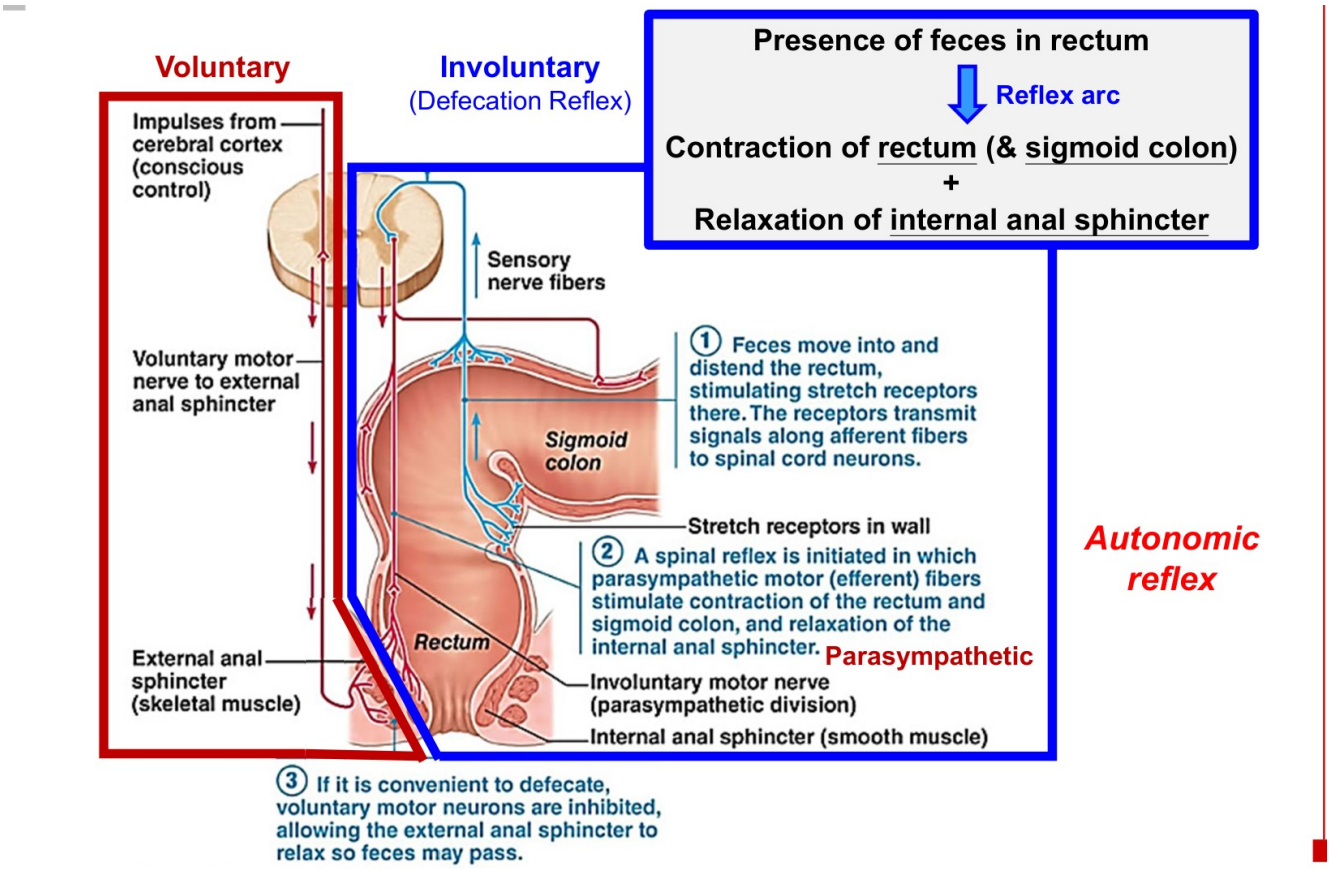
- The circuit is organised locally in the spinal cord, but **normal manifestation requires descending activity**.
- An abnormal adult plantar response helps identify a **corticospinal tract lesion**.



Joseph Jules François Félix Babinski • plantar response

11 Defecation reflex

Faeces enter and distend the rectum → **rectal stretch receptors**
→ sensory afferents → **spinal cord** → parasympathetic efferents
→ **rectum + sigmoid colon contract**
+ **internal anal sphincter relaxes**

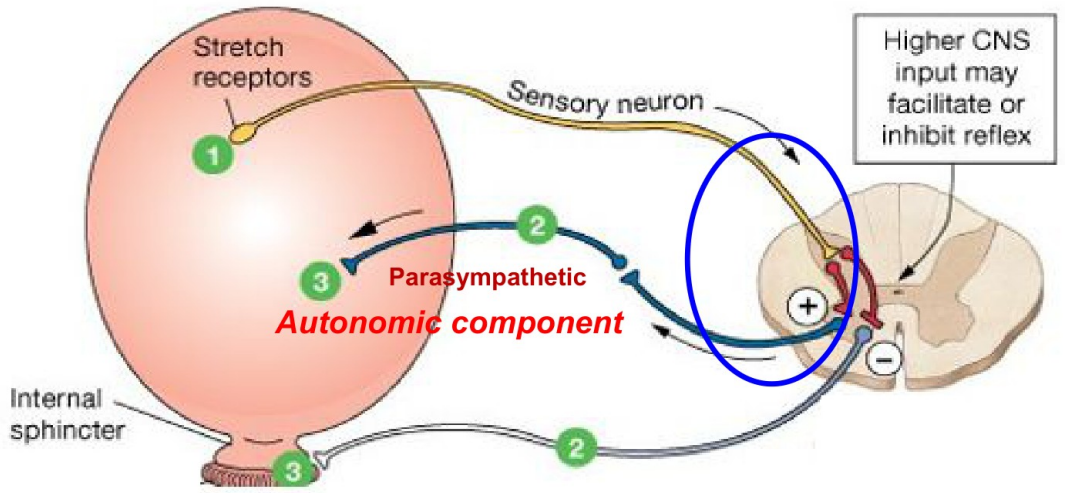


Component	Muscle / control	Action
Internal anal sphincter	Smooth muscle; autonomic	Relaxes as part of the involuntary defecation response.
External anal sphincter	Skeletal muscle; somatic / voluntary	Conscious control can postpone passage. When appropriate, motor activity is reduced → sphincter relaxes → faeces pass.

- **Spinal autonomic reflex + voluntary control of the external sphincter** coordinate the final action.

12 Micturition reflex

Urine fills and distends the bladder → **bladder stretch receptors**
→ sensory afferents → **spinal cord** → parasympathetic efferents
→ **bladder wall contracts + internal urethral sphincter relaxes**
→ promotes bladder emptying



Filling of urine in bladder
↓ **Reflex arc**
Contraction of bladder
+
Relaxation of internal urethral sphincter

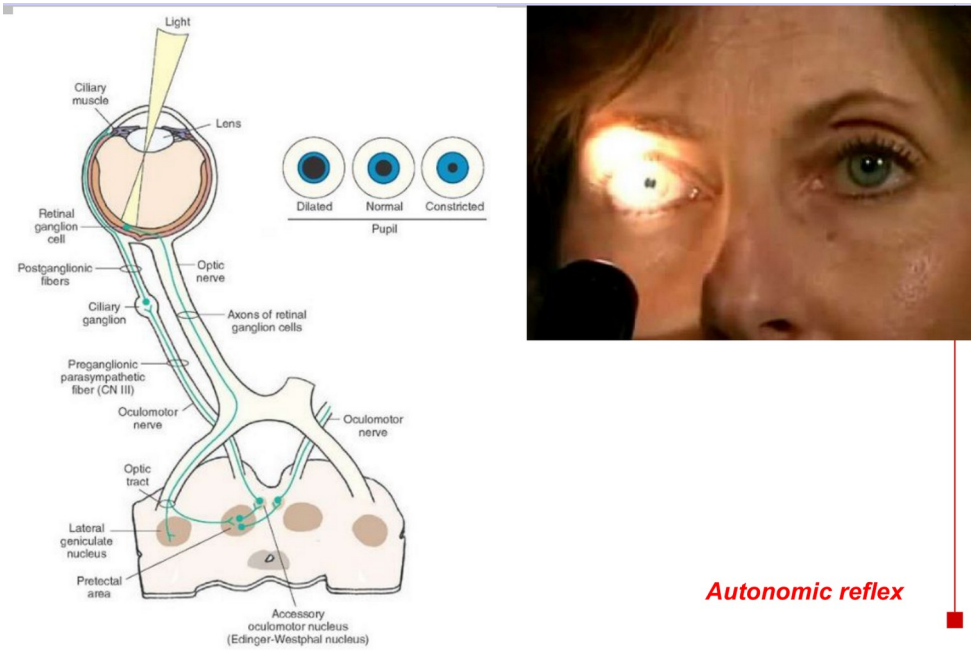
- Higher CNS input can **facilitate or inhibit** the reflex.
- Bladder contraction and outlet relaxation must be coordinated for effective emptying.

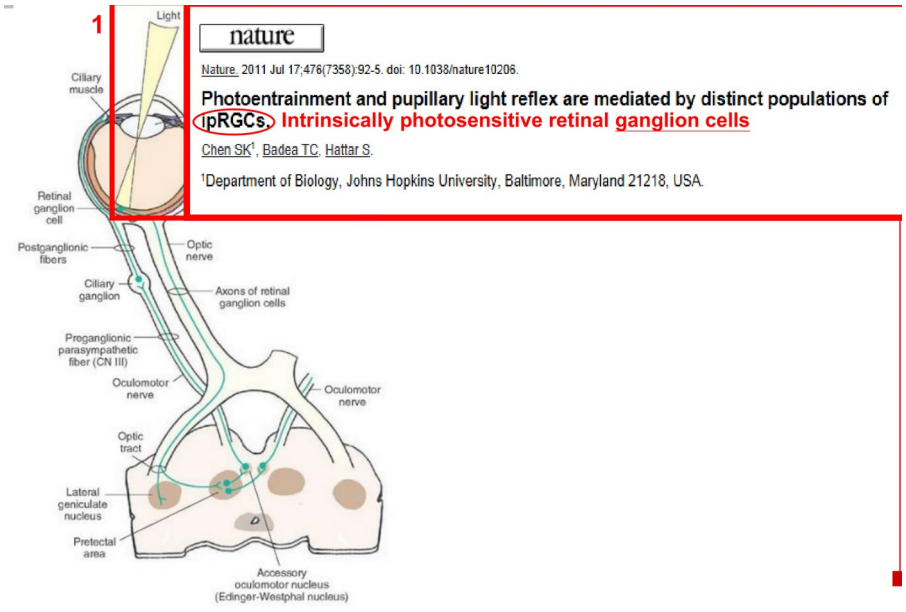
Shared feature	Defecation	Micturition
Filling stimulus	Faeces distend rectum	Urine distends bladder
Receptor / centre	Stretch receptor / spinal cord	Stretch receptor / spinal cord
Autonomic output	Parasympathetic	Parasympathetic
Visceral response	Rectum and sigmoid contract	Bladder contracts
Internal outlet	Anal sphincter relaxes	Urethral sphincter relaxes

Distension → spinal parasympathetic reflex → organ contraction + internal outlet relaxation.

13 Pupillary light reflex: sensory input

Light → **retinal photoreceptive system**
→ **optic nerve (CN II)** → midbrain integration
→ parasympathetic output through **oculomotor nerve (CN III)**
→ circular pupillary constrictor contracts → **pupil size ↓**

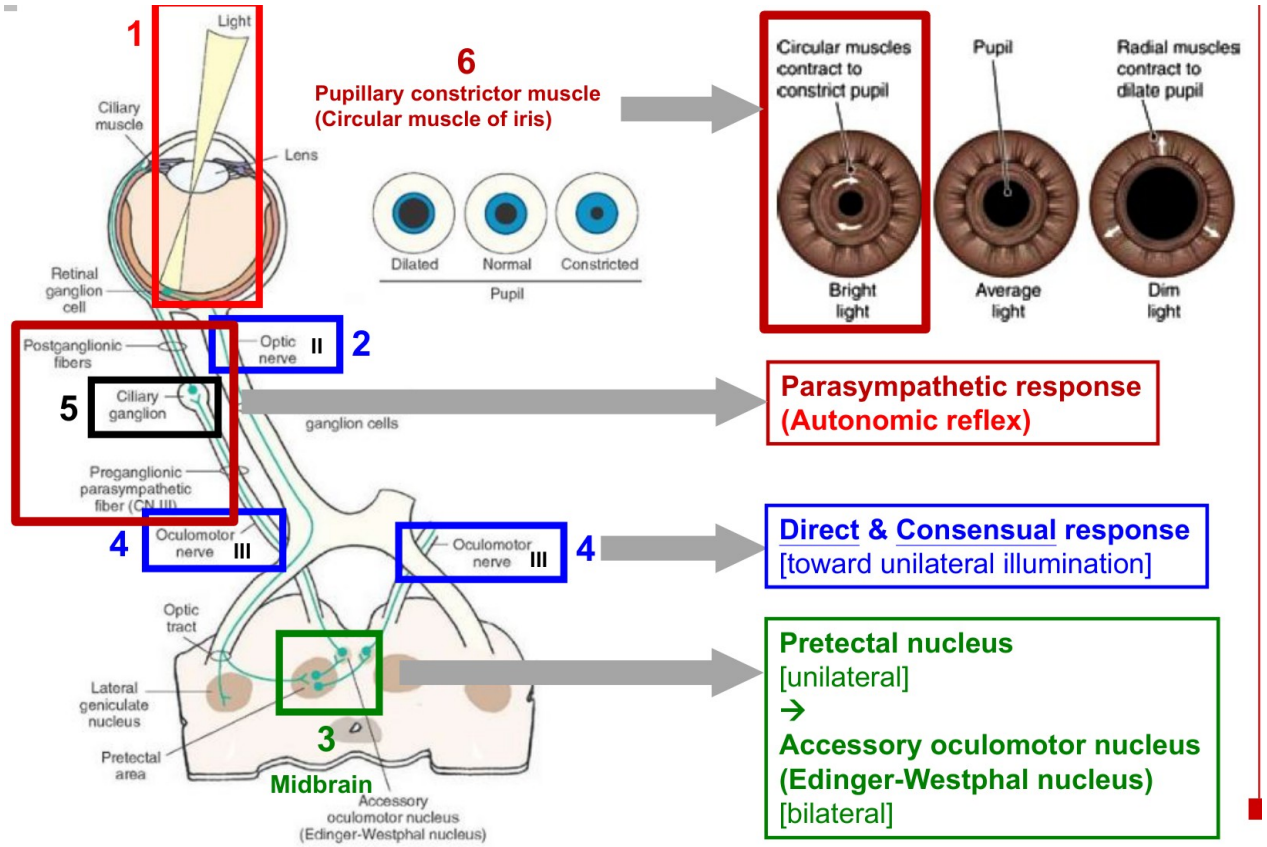




- **Intrinsically photosensitive retinal ganglion cells (ipRGCs)** contribute to the light signal controlling the pupil.
- The effector is the **pupillary constrictor (sphincter pupillae), the circular muscle of the iris**.
- Circular muscle contraction constricts the pupil; radial muscle contraction dilates it.

14 Pupillary circuit: why both eyes respond

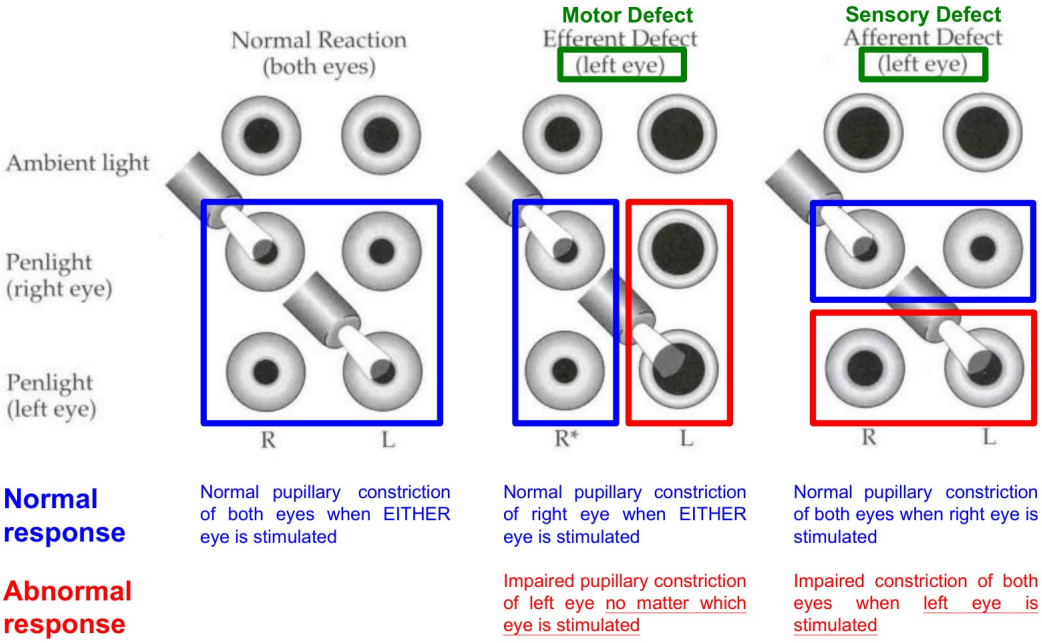
Light in one eye → retina → **CN II** → optic chiasm / optic tract
→ **pretectal nucleus in midbrain**
→ **BOTH Edinger–Westphal nuclei** (accessory oculomotor nuclei)
→ each **CN III** carries preganglionic parasympathetic fibres
→ **ciliary ganglion** → postganglionic fibres through short ciliary nerves
→ **sphincter pupillae contracts in both eyes**



Response	Meaning
Direct response	Constriction of the illuminated eye.
Consensual response	Constriction of the opposite, non-illuminated eye.

- The **bilateral projection to Edinger–Westphal nuclei** explains the paired response to unilateral illumination.
- **CN II is the afferent limb; CN III is the efferent parasympathetic limb.** The ciliary ganglion is the peripheral relay before the iris.

15 Pupillary defects: test each eye



Condition	Light in RIGHT eye	Light in LEFT eye
Normal	Both pupils constrict.	Both pupils constrict.
LEFT sensory / afferent defect	Both pupils constrict: right input reaches both motor pathways.	Constriction of **both** pupils is impaired: left sensory input is defective.
LEFT motor / efferent defect	Right pupil constricts; **left constriction is impaired**.	Right pupil constricts; **left constriction is impaired**.

Reason through the circuit

- Afferent defect: the effect depends on **which eye receives the light**. Input from the affected eye cannot adequately drive either side.
- Efferent defect: the effect depends on **which pupil must constrict**. The affected pupil responds poorly regardless of the illuminated eye.

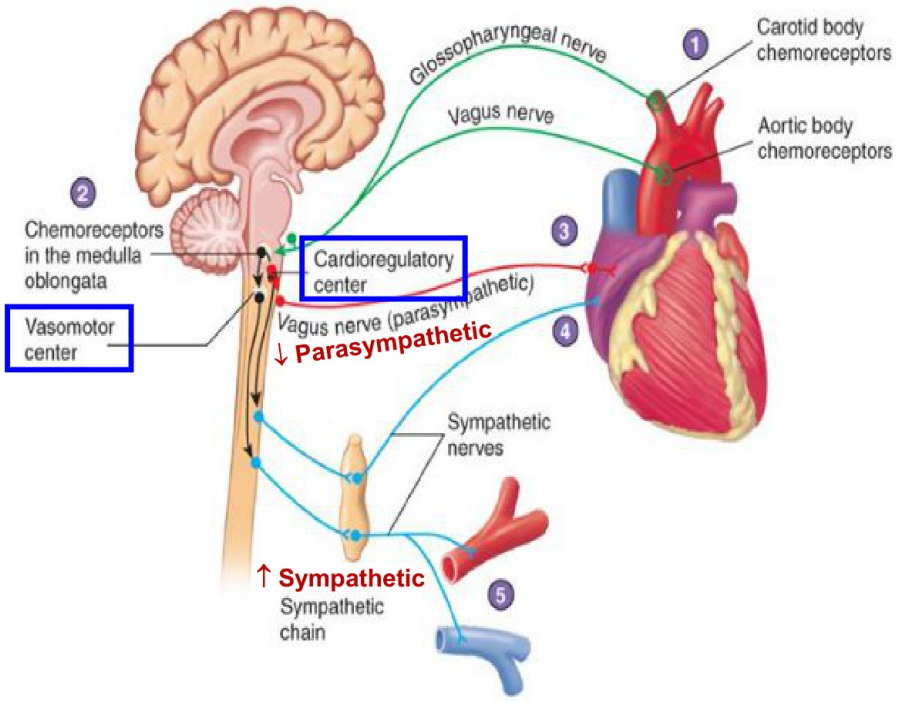
Both pupils fail when one eye is illuminated → suspect that eye's sensory limb.
One pupil fails whichever eye is illuminated → suspect that pupil's motor limb.

These patterns describe an isolated left-sided defect; a partial lesion may impair rather than completely abolish constriction.

16 Chemoreceptor reflex

Blood CO_2 ↑ / O_2 ↓ / pH ↓ → chemoreceptor activation
→ carotid body via **glossopharyngeal nerve (CN IX)**
+ aortic body via **vagus nerve (CN X)**
→ **medullary cardiovascular centres**
→ **parasympathetic activity ↓ + sympathetic activity ↑**
→ **heart rate ↑ + contraction force ↑ + vasoconstriction**

1. Chemoreceptors in the carotid and aortic bodies monitor blood O_2, CO_2, and pH.
2. Chemoreceptors in the medulla oblongata monitor blood CO_2 and pH.
3. Decreased blood O_2, increased CO_2, and decreased pH decrease parasympathetic stimulation of the heart, which increases the heart rate.
4. Decreased blood O_2, increased CO_2, and decreased pH increase sympathetic stimulation of the heart, which increases the heart rate and force of contraction.
5. Decreased blood O_2, increased CO_2, and decreased pH increase sympathetic stimulation of blood vessels, which increases vasoconstriction.

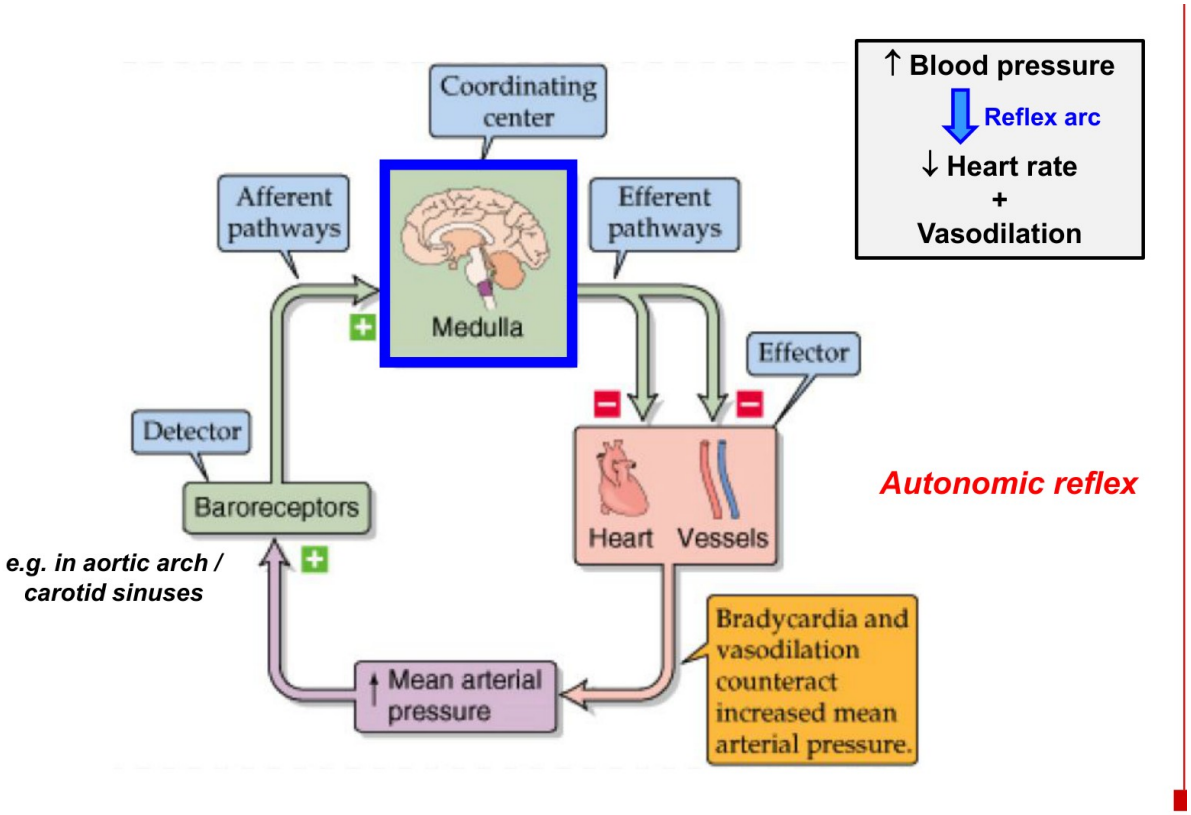


Sensor / centre	Key distinction
Peripheral chemoreceptors	Carotid and aortic **bodies** detect changes in blood gases / pH.
Central chemoreceptors	Located in the medulla; respond mainly to CO_2-driven changes in local H^+ / pH.
Medullary output	Cardioregulatory centre controls cardiac autonomic activity; vasomotor centre controls vascular sympathetic activity.

Body = chemical sensing; carotid sinus / aortic arch = pressure sensing. Cardiovascular and respiratory responses together support blood-gas homeostasis.

17 Baroreceptor reflex: negative feedback

Blood pressure ↑ → arterial wall stretch ↑ → baroreceptor firing ↑
→ carotid sinus via **CN IX** + aortic arch via **CN X**
→ **medulla** → **parasympathetic activity ↑ + sympathetic activity ↓**
→ **heart rate ↓ + vasodilation** → **blood pressure ↓ toward normal**



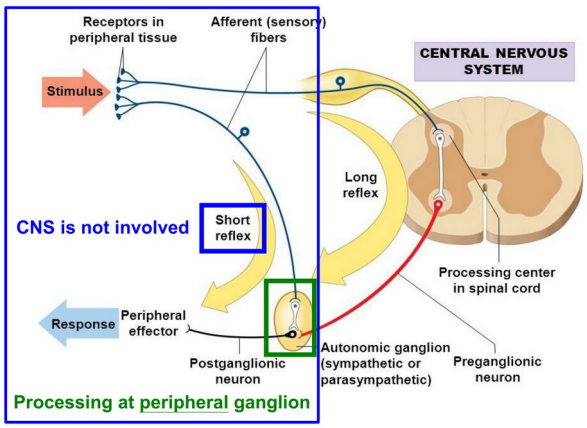
Negative feedback: pressure returns toward normal
→ excessive receptor stretch / afferent firing subsides
→ the corrective autonomic response diminishes.

Compare	Chemoreceptor reflex	Baroreceptor reflex (BP rise)
Initial stimulus	CO_2 ↑ / O_2 ↓ / pH ↓	Blood pressure / stretch ↑
Autonomic change	Parasympathetic ↓; sympathetic ↑	Parasympathetic ↑; sympathetic ↓
Cardiovascular effect	Heart rate ↑; vasoconstriction	Heart rate ↓; vasodilation

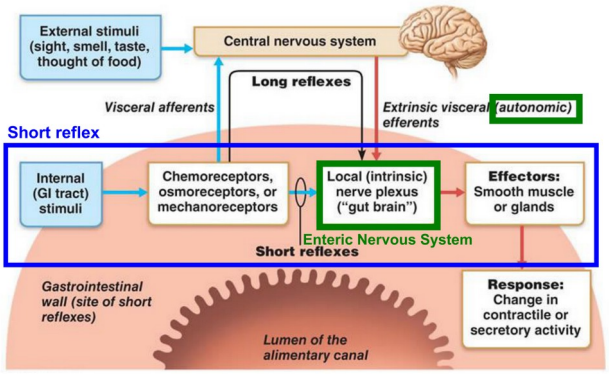
- Vasodilation here follows **reduced sympathetic vasoconstrictor drive**; increased parasympathetic activity slows the heart.

18 Short reflexes & the enteric nervous system

Internal GI stimulus → **chemoreceptor / osmoreceptor / mechanoreceptor**
→ sensory neuron → **local intrinsic nerve plexus ("gut brain")**
→ local motor output → **smooth muscle or gland**
→ contractile or secretory activity changes



Peripheral integration vs a CNS loop



Local enteric control and long reflex inputs

Feature	Short reflex	Long reflex
Integration	**Peripheral ganglia / local nerve plexus**	CNS
CNS required in the reflex arc?	**No**	Yes
GI examples of input	Local mechanical, osmotic or chemical changes	Visceral afferents; external stimuli such as sight, smell, taste or thought of food
Output	Local enteric neurons → smooth muscle / glands	Extrinsic autonomic output → enteric circuits / effectors

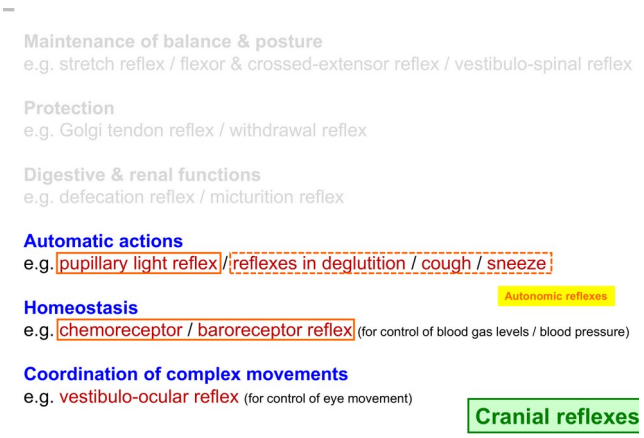
- An enteric short reflex can be completed locally while longer autonomic pathways modulate GI activity.

19 Functions of reflexes

Function	Examples and outcome
Balance & posture	Stretch reflex; flexor and crossed-extensor reflexes; vestibulospinal reflex → maintain position and support.
Protection	Golgi tendon reflex; withdrawal reflex → regulate tension and withdraw from harmful stimulation.
Digestive & renal functions	Defecation reflex; micturition reflex → coordinate visceral emptying.
Automatic actions	Pupillary light reflex; reflexes involved in deglutition (swallowing), coughing and sneezing.
Homeostasis	Chemoreceptor and baroreceptor reflexes → support blood-gas and blood-pressure control.
Coordination of complex movements	Vestibulo-ocular reflex → coordinates eye movement.



Spinal reflex examples



Cranial reflex examples

Keep the classification axes separate

- "Spinal / cranial" describes the **integration site**.
- "Somatic / autonomic" describes the **motor system and effector**.
- A spinal reflex can include autonomic output: defecation and micturition.
- A reflex can be automatic even when its effector is skeletal muscle: stretch and withdrawal reflexes.

20 Integrated pathway revision

Reflex	Receptor → integration → output	Key effect
Stretch	Muscle spindle → Ia → spinal monosynaptic excitation → α motor neuron ↑	Same muscle contracts.
Golgi tendon	Golgi tendon organ → Ib → spinal interneuron circuit → α motor neuron ↓	Same muscle relaxes.
Withdrawal	Pain receptor → spinal polysynaptic circuit → ipsilateral flexor ↑ / extensor ↓	Stimulated limb withdraws.
Crossed extensor	Pain input → crossing spinal interneurons → contralateral extensor ↑ / flexor ↓	Opposite limb supports weight.
Plantar	Cutaneous input → spinal circuit under descending control	Adult toes flex normally.
Defecation	Rectal stretch → spinal cord → parasympathetic output	Rectum / sigmoid contract; internal anal sphincter relaxes.
Micturition	Bladder stretch → spinal cord → parasympathetic output	Bladder contracts; internal urethral sphincter relaxes.
Pupillary light	Retina → II → midbrain → bilateral III / ciliary ganglia	Direct + consensual constriction.
Chemoreceptor	Blood-gas / pH change → IX / X → medulla → autonomic adjustment	Heart rate ↑; vasoconstriction.
Baroreceptor	Pressure ↑ → IX / X → medulla → vagal ↑ / sympathetic ↓	Heart rate ↓; vasodilation.
Enteric short	Local GI receptors → intrinsic plexus → smooth muscle / glands	Local motility / secretion changes.

Six distinctions to retain

- **Ia vs Ib:** spindle stretch input vs tendon tension input.
- **α vs γ:** extrafusal force vs intrafusal sensitivity.
- **Coactivation vs loop:** simultaneous descending drive vs γ → spindle → Ia → α.
- **Ipsilateral vs contralateral:** withdrawal side vs supporting side.
- **Sensory vs motor pupil defect:** illuminated eye determines failure vs affected pupil determines failure.
- **Short vs long:** peripheral integration vs CNS integration.